

02393 Programming in C++

Module 13: Exam Preparation & Conclusion

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Lecture plan

Module	Date	Title	Textbook chapters
0 and 1	03/09	Welcome & C++ Overview	1
2	10/09	Basics of C++ and its Data Types	1.5, 2.1, 2.3 - 2.5, 11.1, 11.3
3	17/09	Memory management	12.1, 11.2, 11.3
4	24/09	LAB DAY	N/A
5	01/10	Libraries and Interfaces	2.2, 2.6 - 2.8, 3.1 - 3.3, 4.1 - 4.3
6	08/10	Classes and Objects	5.1, 6.1, 11.2, 12.1, 12.4, 12.7
Autumn break			
7	22/10	Templates	5.1, 14.2
8	29/10	Inheritance	19.1 - 19.3
9	05/11	LAB DAY	N/A
10	12/11	Recursive Programming	7
11	19/11	Linked Lists	12.2, 12.3
12	26/11	Trees	16.1 - 16.4
13	03/12	LAB DAY	N/A

Outline

- Exam simulation
- The exam
- Lab

Exam simulation

- A new sample exam (December 2022 exam) is now available
 - Hand-out: Course page → Course Content → Content → Exam samples
 - Local autograder: Course page → Course Content → Content → Module 13
- The solution will be available on the next week
- Today you can use this (or other past exams) as an exam simulation
- Make sure you can run the local autograder
- TAs are available to answer your questions on any course topic

The exam

- Date & time: Saturday 21 December 2024, 9:00
- Place: DTU Campus Lyngby, 127/119, 101B/Sportshal 1, 101B/Sportshal 2
 - <http://eksamensplan.dtu.dk/student>
 - <http://eksamensplan.dtu.dk/Course>
- Duration: 4 hours
- All aids allowed **excluding Internet and AI**
- Marking: pass/fail

The exam

- The exam contains various exercises with multiple tasks
 - Each exercise has program `exZZ-main.cpp` and a header `exZZ-library.h`
 - For each of the tasks (a), (b), . . .
 - You are asked to implement code in `exZZ-library.cpp`
 - You might be asked to also modify `exZZ-library.h`
 - You get points for each task you complete, in any order
 - Some tasks may depend on others (if so, it is written)
 - You can use `exZZ-main.cpp` to check your solutions
 - During the exam you can run the autograding scripts locally on your computer
 - Autolab is **NOT available**
 - Make sure you have your computer properly configured before the exam
 - We might run more tests after the exam to grade your solutions

The exam

- Some topics you can expect:
 - Iterate over arrays/vectors/matrices/. . .
 - Some list-like or tree-like structure
 - Implement one or more recursive functions
 - You can choose to avoid using recursion, but it may be harder!
 - Deal with pointers
 - Basic use of STL containers (vectors, sets, maps, . . .)
 - Declare a class and/or implement its methods
 - Extend a class using inheritance
 - Implement (part of) a parametric data structure using templates
 - ... (other topics seen during the course)

The exam

- Submission:
 - Electronic submission on DTU Digital Eksamen: <https://eksamen.dtu.dk>
 - Please remember to submit your final solution to DTU Digital Eksamen!
 - Otherwise, it will count as a blank exercise
 - You can submit your solution only once
 - Make sure to upload the correct files as specified in the hand-out
 - Only one file will be marked as main document, while the others be marked as appendixes
 - Don't worry, all files will be considered for the assessment (as long as you uploaded them).

Thank you for your active participation!

Best of luck with the exam!

Lab time

- Today
 - Make sure C++ works on your computer, request help if it doesn't
 - Try to solve the exam sample from December 2022 available on DTU Learn
 - Ask questions if something is unclear (including previous assignments)

DTU

